

CHAPTER 6

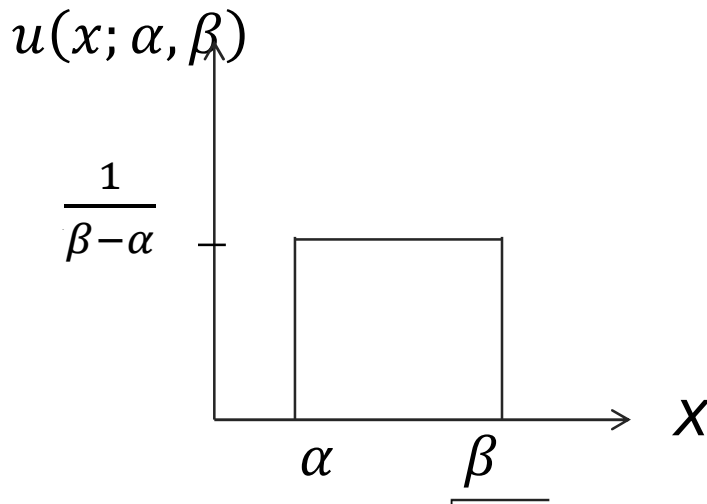
In this chapter, discuss few widely used continuous probability distributions.

6.1 Uniform Distribution

Definition 6.1: A random variable has a uniform distribution and it is referred to as a continuous uniform random variable if and only if its probability density is given by,

$$u(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta - \alpha} & \text{for } \alpha < x < \beta \\ 0 & \text{elsewhere} \end{cases}$$

The parameters α and β of this probability density are real constants with $\alpha < \beta$



Theorem 6.1: The mean and the variance of the uniform distribution are given by,

$$\mu = \frac{\alpha + \beta}{2} \text{ and } \sigma^2 = \frac{1}{12}(\beta - \alpha)^2$$

Proof:

$$\begin{aligned}
 \mu = E[X] &= \int_{\alpha}^{\beta} x \cdot f(x) dx \\
 &= \int_{\alpha}^{\beta} x \cdot \frac{1}{\beta - \alpha} dx \\
 &= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} x \cdot dx \\
 &= \frac{1}{\beta - \alpha} \left[\frac{x^2}{2} \right]_{\alpha}^{\beta} \\
 &= \frac{1}{(\beta - \alpha)} \cdot \frac{\beta^2 - \alpha^2}{2} \\
 &= \frac{1}{(\beta - \alpha)} \cdot \frac{(\beta - \alpha)(\beta + \alpha)}{2} \\
 &= \frac{(\alpha + \beta)}{2}
 \end{aligned}$$

$$\begin{aligned}
 \sigma^2 &= \int_{\alpha}^{\beta} (x - \mu)^2 \cdot f(x) dx \\
 &= \int_{\alpha}^{\beta} \left[x - \frac{(\alpha + \beta)}{2} \right]^2 \frac{1}{(\beta - \alpha)} dx
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{(\beta - \alpha)} \int_{\alpha}^{\beta} \left[x^2 - 2 \frac{(x + \beta)}{2} \cdot x + \frac{(\alpha + \beta)^2}{4} \right] \\
&= \frac{1}{(\beta - \alpha)} \left[\frac{x^3}{3} - \frac{(\alpha + \beta)x^2}{2} + \frac{(\alpha + \beta)^2}{4} x \right]_{\alpha}^{\beta} \\
&= \frac{1}{(\beta - \alpha)} \left[\frac{(\beta^3 - \alpha^3)}{3} - \frac{(\alpha + \beta)(\beta^2 - \alpha^2)}{2} \right. \\
&\quad \left. + \frac{(\alpha + \beta)^2(\beta - \alpha)}{4} \right] \\
&= \frac{1}{(\beta - \alpha)} \left[\frac{(\beta - \alpha)(\beta^2 + \alpha\beta + \alpha^2)}{3} \right. \\
&\quad \left. - \frac{(\alpha + \beta)(\beta + \alpha)(\beta - \alpha)}{2} + \frac{(\alpha + \beta)^2(\beta - \alpha)}{4} \right] \\
&= \frac{(\beta^2 + \alpha\beta + \alpha^2)}{3} - \frac{(\alpha + \beta)^2}{2} + \frac{(\alpha + \beta)^2}{4} \\
&= \frac{(\beta^2 + \alpha\beta + \alpha^2)}{3} - \frac{\alpha^2 + 2\alpha\beta + \beta^2}{4} \\
&= \frac{1}{12} (\beta^2 - 2\alpha\beta + \alpha^2) \\
&= \frac{1}{12} (\beta - \alpha)^2
\end{aligned}$$

The CDF of Uniform Distribution

Example 6.1

The waiting time, X , in minutes for a customer to receive a service at a bank is uniformly distributed between 5 and 15 minutes.

- (a) Find the probability that a randomly selected customer will wait less than 8 minutes.
- (b) Find the probability that a customer waits between 7 and 12 minutes.
- (c) Find the probability that a customer waits more than 13 minutes.

Example 6.2

The time, X , in minutes taken by a student to complete a particular statistics exercise is uniformly distributed between 10 and 30 minutes.

- (a) Find the probability that a student takes between 15 and 25 minutes.
- (b) Find the value of k such that $P(X > k) = 0.2$
- (c) Find the expected completion time and the variance

6.2 Gamma Distribution

Definition 6.2: A random variable X has a gamma distribution and it is referred to as a gamma random variable if and only if its probability density is given by

$$g(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$

Properties of Gamma Function

Theorem 6.2 The r^{th} moment about the origin of the gamma distribution is given by,

$$\mu_r' = \frac{\beta^r \Gamma(\alpha + r)}{\Gamma(\alpha)}$$

Proof:

$$\begin{aligned}\mu_r' = E[x^r] &= \int_0^\infty x^r \cdot \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} dx \\ &= \frac{\beta^r}{\Gamma(\alpha)} \cdot \int_0^\infty y^{\alpha+r-1} e^{-y} dy\end{aligned}$$

where we let $y = \frac{x}{\beta}$.

Since the integral on the right is $\Gamma(r + \alpha)$ according to the definition of the gamma function,

$$\mu_r' = \frac{\beta^r \Gamma(\alpha + r)}{\Gamma(\alpha)}$$

Theorem 6.3: The mean and the variance of the gamma distribution are given by,

$$\mu = \alpha\beta \text{ and } \sigma^2 = \alpha\beta^2$$

Exercise: Prove above theorem

(Hint: use theorem 6.2)

Applications:

Suppose that Poisson events are occurring at the constant rate of λ per unit time. Let the random variable Y denote the **waiting time** for the r^{th} event. Then Y has a gamma distribution with parameters r and λ .

$$f_Y(y) = \frac{\lambda^r}{\Gamma(r)} y^{r-1} e^{-\lambda y}, y > 0 \text{ where}$$

$$r = \alpha \text{ and } \lambda = \frac{1}{\beta}$$

Theorem 6.4: The moment generating function of the gamma distribution is given by,

$$M_X(t) = (1 - \beta t)^{-\alpha}$$

Applications:

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Theorem 6.4: Then moment generating function of the gamma distribution is given by,

$$M_X(t) = (1 - \beta t)^{-\alpha}$$

6.3 Exponential Distribution

Exponential distribution is a special case of the gamma distribution (where $\alpha=1$ and $\beta=\theta$) that plays important role in Statistics.

Definition 6.3: A random variable X has an exponential distribution and it is referred to as an exponential random variable if and only if its probability density is given by

$$g(x; \theta) = \begin{cases} \frac{1}{\theta} e^{-x/\theta} & \text{for } x > 0 \\ 0 & \text{elsewhere} \end{cases} \quad \text{where } \theta > 0$$

Corollary to the theorem 6.3 the mean and the variance of the exponential distribution are given by

$$\mu = \theta \text{ and } \sigma^2 = \theta^2$$

Proof:

CDF of Exponential Distribution

Applications:

1. Suppose that Poisson events are occurring at the constant rate of λ per unit time. Let the random variable Y denote the **waiting time** for the **first event**. Then Y has an exponential distribution with parameter θ .

2. Also the exponential distribution can be considered as a continuous counterpart of the geometric distribution. (describes the time for a continuous process to change its state)

Eg. Time for an electric bulb to fail (life time of an electric bulb)

3. Also it applies to the waiting time between events.

Exercise: A certain location on a highway, the number of cars exceeding the speed limit by more than 40 km/h in half an hour is a random variable having Poisson distribution with $\lambda = 8.4$. What is the probability of a waiting time of less than 5 minutes between cars exceeding the speed limit by more than 40km/h?

6.4 Normal Distribution

By far, the most widely used probability model in Statistic is the normal (or Gaussian) distribution.

Definition 6.4: A random variable X has a normal distribution and it is referred to as a normal random variable if and only if its probability density is given by,

$$N(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty \quad \text{where} \\ \sigma > 0$$

The graph of the normal distribution, shaped like a cross section of a bell.

Let us show that the formula of definition 6.4 can serve as a probability density.

Since $\sigma > 0$, the values of $N(x; \mu, \sigma)$ are positive. Hence it satisfies the first principle of a density function.

Then we must show that the total area under the curve is equal to 1.

Integrating from $-\infty$ to ∞ and making substituting $z = \frac{x-\mu}{\sigma}$, we get

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz \\ &= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{1}{2}z^2} dz\end{aligned}$$

By substituting $\frac{z^2}{2} = y$, in the definition of the gamma function, it can be shown that,

$$\int_0^{\infty} e^{-\frac{1}{2}z^2} dz = \frac{\Gamma\left(\frac{1}{2}\right)}{\sqrt{2}} = \frac{\sqrt{\pi}}{\sqrt{2}}$$

Theorem 6.4: The moment generating function of the normal distribution is given by

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

Proof: By definition,

$$\begin{aligned}M_X(t) &= \int_{-\infty}^{\infty} e^{xt} \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}[-2xt\sigma^2 + (x-\mu)^2]} dx\end{aligned}$$

By rearranging $-2xt\sigma^2 + (x - \mu)^2$

$$\begin{aligned}
& -2xt\sigma^2 + (x - \mu)^2 \\
& = [x - (\mu + t\sigma^2)]^2 - 2\mu t\sigma^2 - t^2\sigma^4
\end{aligned}$$

Hence we get,

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \left\{ \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[\frac{x - (\mu + t\sigma^2)}{\sigma}\right]^2} dx \right\}$$

Since the quantity inside the braces is the integral from $-\infty$ to ∞ of a normal density with the parameter $\mu + t\sigma^2$ and σ , and hence it is equal to 1.

It follows that,

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

Exercise: Verify that the parameter μ and σ in definition 6.4 are, indeed, mean and standard deviation of the normal distribution.

Definition 6.5: The normal distribution with $\mu = 0$ and $\sigma = 1$ is referred to as the **standard normal distribution**.

Theorem 6.5: If X has a normal distribution with the mean μ and standard deviation σ , then, $Z = \frac{X - \mu}{\sigma}$ has the standard normal distribution.

Proof: Since the relationship between the values of X and Z is linear, Z must take on a value between $z_1 = \frac{x_1 - \mu}{\sigma}$ and $z_2 = \frac{x_2 - \mu}{\sigma}$ when X takes on a value between x_1 and x_2 . Hence

$$\begin{aligned}
 Pr(x_1 < X < x_2) &= \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
 &= \frac{1}{\sigma\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{1}{2}z^2} dz \\
 &= \int_{z_1}^{z_2} n(z; 0,1) dz \\
 &= Pr(z_1 < Z < z_2)
 \end{aligned}$$

Where Z is seen to be a random variable having the standard normal distribution.

Thus to use the standard normal table in connection with any random variable having a normal distribution, we simply perform the change of scale $z = \frac{x-\mu}{\sigma}$

The Normal Approximation to the Binomial Distribution

The normal distribution is sometimes introduced as a continuous distribution that provides a close approximation to the binomial distribution when n , the number of trials, is

very large and θ , the probability of a success on an individual trial, is close to 0.5.

Steps to working a normal approximation to the binomial distribution

1. Identify success, the probability of success, the number of trials, and the desired number of successes. Since this is a binomial problem, these are the same things which were identified when working a binomial problem.

2. Convert the discrete x to a continuous x .

$$Pr(X \leq x) = Pr(X \leq x + 0.5) \text{ or}$$

$$Pr(X \geq x) = Pr(X \geq x - 0.5)$$

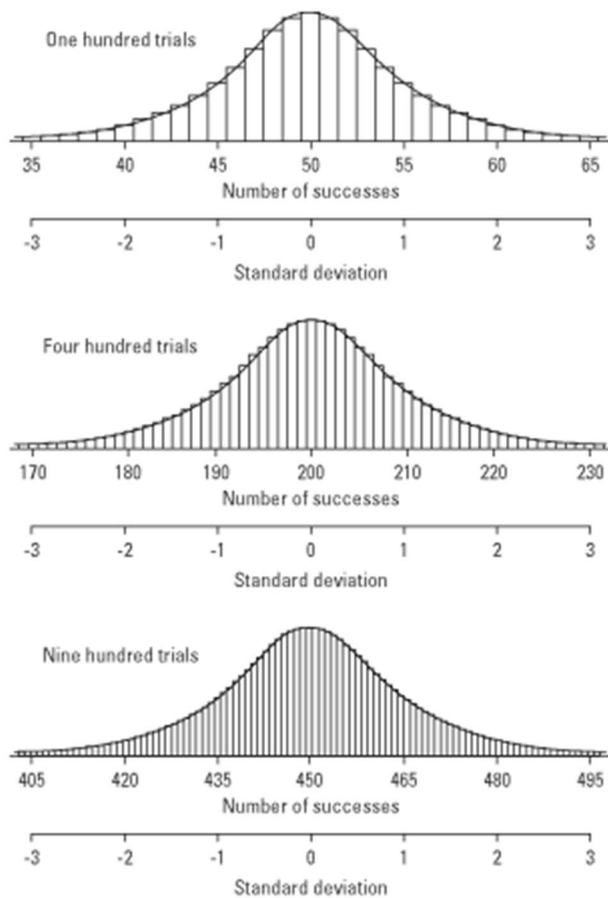
3. Find the mean, $\mu = n\theta$

4. Find the standard deviation, $\sigma = \sqrt{n\theta(1 - \theta)}$

5. Compute the z-score using the standard formula for an individual score.

$$Z = \frac{X - n\theta}{\sqrt{n\theta(1 - \theta)}}$$

6. Calculate the probability desired.



Note: Strictly speaking, our results apply when $n \rightarrow \infty$, but the normal distribution is often used to approximate binomial probabilities even when n is fairly small. A good rule of thumb is to use this approximation only when $n\theta$ and $n(1-\theta)$ are both greater than 5.

Exercise: Use the normal approximation to the binomial distribution to determine the probability of getting 6 heads and 10 tails in 16 flips of a balanced coin.

Solution: Let X is number of heads from 16 flipping of a balanced coin.

X has a binomial distribution with $n=16$ and $\theta=0.5$. Hence the required probability is $\Pr(X=6)$.

To approximate this probability using normal distribution, it is necessary to perform continuity correction. i.e. $\Pr(X=6)=\Pr(5.5<X<6.5)$.

Then the required probability can be evaluated using normal distribution with mean $\mu = n\theta = 16 \times 0.5 = 8$ and standard deviation $\sigma = \sqrt{n\theta(1 - \theta)} = \sqrt{16 \cdot 0.5 \cdot 0.5} = 2$.

The Normal Approximation to the Poisson Distribution

For sufficiently large values of $\lambda = \alpha t$ (say $\lambda > 20$) the normal distribution with mean $\mu = \lambda$ and variance $\sigma^2 = \lambda$, is an excellent approximation to the Poisson distribution if an appropriate continuity correction is done.

i.e. $\Pr(X \leq x) = \Pr(X \leq x + 0.5)$ or
 $\Pr(X \geq x) = \Pr(X \geq x - 0.5)$

Exercise: A cable manufacturer has determined that for a 4-meter unit of cable there are $\alpha=4$ defects per 4-meter. Determine the probability of getting 7 defects in $t=1.5$ units of cable using normal appropriation.

Solution:

Let X is number of defects per unit are.

X has a Poisson distribution with parameter

$$\lambda = \alpha t = (4 \text{ defects per 4-meters}) \times (1.5 \text{ units}) = 6.0$$

This can be appropriated by using normal distribution with parameters $\mu = \lambda = 6.0$ and standard deviation $\sigma = \sqrt{\lambda} = \sqrt{6.0} = 2.4495$.

The required probability is,
 $\Pr(6.5 < X < 7.5)$ in Poisson distribution.